ClipSAT : ACT Math

Unit 0.01: ACT Math Diagnostic & Pacing Strategy Drill

Name: _____

Q1. For the imaginary unit i , where i^2 = -1 , which of the following complex numbers is equal to 5 - 3i - 2 - 8i ?

[ ] A: 3 + 5i
[ ] B: 3 - 11i
[ ] C: 7 - 11i
[ ] D: 3 - 5i

Q2. The bar graph below shows the number of hours four students volunteered at a community center last week. What percent of the total volunteer hours logged by the four students was logged by Maya?

[ ] A: 14%
[ ] B: 28%
[ ] C: 35%
[ ] D: 42%

Q3. In triangle ABC shown below, the measure of angle A is 52^circ and the measure of angle B is 68^circ . What is the value of x ?

[ ] A: 58
[ ] B: 60
[ ] C: 70
[ ] D: 120

Q4. In the right triangle triangle ABC shown below, the right angle is at vertex C. The lengths of the sides are given in inches: AC = 12, BC = 5, and AB = 13. What is the value of cos B?

[ ] A: frac{5}{13}
[ ] B: frac{12}{13}
[ ] C: frac{5}{12}
[ ] D: frac{12}{5}

Q5. For the imaginary unit i , where i^2 = -1 , which of the following complex numbers is equal to 5 - 3i - 2 - 8i ?

[ ] A: 3 + 5i
[ ] B: 3 - 11i
[ ] C: 7 - 11i
[ ] D: 3 - 5i

Q6. Which of the following expressions is equivalent to 5x^2 - 2x + 7 - 3x^2 - 6x - 4 ?

[ ] A: 2x^2 - 8x + 3
[ ] B: 2x^2 - 8x + 11
[ ] C: 2x^2 + 4x + 3
[ ] D: 2x^2 + 4x + 11

Q7. In the standard x, y coordinate plane, what are the coordinates of the midpoint of a line segment with endpoints -3, 8 and 5, -2?

[ ] A: 1, 3
[ ] B: 4, -5
[ ] C: 2, 6
[ ] D: 3, 1

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Q8. If fx = 3x^2 - 5x + 4 , what is the value of f-2 ?

[ ] A: -18

[ ] B: 2

[ ] C: 6

[ ] D: 26